Online Machine Learning

Online Dimension Reduction via Sliced Inverse Regression

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Introduction and Motivation

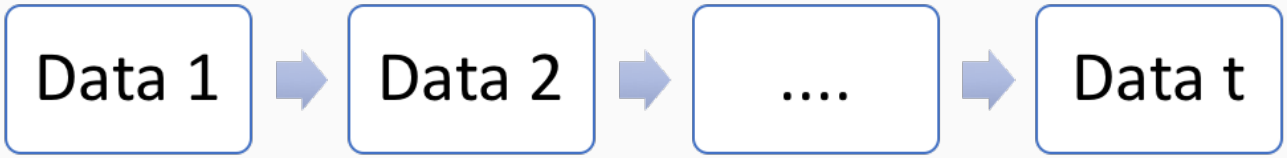
We are concerned with machine learning with large data sets.
When the data sets are too large, many common approaches fail.
Simply because they cannot load the data into memory or they are not efficient enough.
Two common approaches:

- Parallelized machine learning
- Online machine learning

Why Online Learning

Traditional batch learning algorithms deal with the whole data set:

Online learning only requires small data blocks or instances divided from the data set:



Concerns With Online Learning

Advantages

(a) Fast Computation

(b) Low Memory Requirement

Disadvantages

(a) Slow Convergence

(b) Lack of Theoretical Guarantees

Existing Online Learning Methods

Online Regularization Method(LASSO) (Xiao [3])
Online Principal Component Analysis (Oja and Karhunen [2])
Online Dictionary Learning (Mairal et al. [1])

Online Regularization Method (LASSO)

Consider the objective function

$$E\{L(\beta, Z)\} + P_\lambda(\beta)$$

where $Z = (X, Y)$ is the data.
For $t = 1, 2, \ldots,$ denote $Z_1, Z_2, \ldots, Z_t$ as the incoming data,

Online Regularization Method (LASSO)

For $t = 1, 2, \ldots,$ do

1. Given the function $L_t(\beta_{t-1}, Z_t)$, compute the subgradient $g_t = \partial L_t$.
2. Update the average subgradient

$$\bar{g}_t = \frac{t-1}{t}\bar{g}_{t-1} + \frac{1}{t}g_t$$

3. Compute the next β_t

$$\beta_{t+1} = \text{argmin}\{\bar{g}_t{}^\top \beta + P_\lambda(\beta) + h(\beta)/\sqrt{t}\}$$

where $h(\beta)$ is a strongly convex function.

Online Principal Component Analysis

We notice that, seeking the K principal eigenvectors of $\widehat{\mathbf{M}}_{t+1}$ amounts to

$$\begin{aligned} \underset{\mathbf{B}}{\text{maxmize}} \quad & \text{tr}(\widehat{\mathbf{M}}_{t+1}\mathbf{B}\mathbf{B}^\top) \\ \text{subject to} \quad & \mathbf{B}^\top\mathbf{B} = \mathbf{I}_{K\times K} \end{aligned} \tag{1}$$

The first order derivative of the objective function is $2\widehat{\mathbf{M}}_{t+1}\mathbf{B}$, which inspires the proposal of the following online stochastic gradient descent optimization:

$$\widehat{\mathbf{B}}_{t+1} = \mathcal{P}_{\text{orth}}(\widehat{\mathbf{B}}_t + \gamma_t\widehat{\mathbf{M}}_{t+1}\widehat{\mathbf{B}}_t), \tag{2}$$

where γ_t is a small step-size.

Online Dictionary Learning

Classical dictionary learning:

$$f_n(D) \stackrel{\text{def}}{=} \sum_{i=1}^{n} l(x_i, D)$$

where $x_i \in \mathbb{R}^m$ is the sample. $D \in \mathbb{R}^{K \times 1}$ is the dictionary, each column representing a basis vector. l is a loss function such that $l(x, D)$ should be small if D is good at representing the signal x in a sparse fashion.

Online Dictionary Learning

Algorithm 1 Online dictionary learning.

Require: $\mathbf{x} \in \mathbb{R}^m \sim p(\mathbf{x})$ (random variable and an algorithm to draw i.i.d samples of p), $\lambda \in \mathbb{R}$ (regularization parameter), $\mathbf{D}_0 \in \mathbb{R}^{m \times k}$ (initial dictionary), T (number of iterations).

$\mathbf{A}_0 \in \mathbb{R}^{k \times k} \leftarrow 0$, $\mathbf{B}_0 \in \mathbb{R}^{m \times k} \leftarrow 0$ (reset the "past" information).

for $t = 1$ to T **do**

  Draw $\mathbf{x}_t$ from $p(\mathbf{x})$.

  Sparse coding: compute using LARS

$$\alpha_t \triangleq \underset{\alpha \in \mathbb{R}^k}{\arg\min} \frac{1}{2} ||\mathbf{x}_t - \mathbf{D}_{t-1}\alpha||_2^2 + \lambda ||\alpha||_1.$$

  $\mathbf{A}_t \leftarrow \mathbf{A}_{t-1} + \alpha_t \alpha_t^T$.

  $\mathbf{B}_t \leftarrow \mathbf{B}_{t-1} + \mathbf{x}_t \alpha_t^T$.

  Compute $\mathbf{D}_t$ using Algorithm 2, with $\mathbf{D}_{t-1}$ as warm restart, so that

$$\begin{aligned} \mathbf{D}_t &\triangleq \underset{\mathbf{D} \in \mathcal{C}}{\arg\min} \frac{1}{t} \sum_{i=1}^{t} \left(\frac{1}{2} ||\mathbf{x}_i - \mathbf{D}\alpha_i||_2^2 + \lambda ||\alpha_i||_1 \right), \\ &= \underset{\mathbf{D} \in \mathcal{C}}{\arg\min} \frac{1}{t} \left(\frac{1}{2} \mathrm{Tr}(\mathbf{D}^T \mathbf{D} \mathbf{A}_t) - \mathrm{Tr}(\mathbf{D}^T \mathbf{B}_t) \right). \end{aligned} \tag{6}$$

end for

Return $\mathbf{D}_T$ (learned dictionary).

Main Interests

Convergence properties: almost sure convergence (stochastic process), exact convergence rate.

Regret bound: defined as the difference of the two loss functions:

$$R(\beta) \stackrel{\text{def}}{=} \sum_{t=1}^{T} L_t(\beta_t) - \sum_{t=1}^{T} L_t(\beta)$$

Introduction and Motivation

- Sufficient dimension reduction seeks to find a matrix $\mathbf{B} \in \mathbb{R}^{p \times K}$ such that

$$Y \perp\!\!\!\perp \mathbf{x} \mid (\mathbf{B}^\top \mathbf{x})$$

- Since SIR (Li, 1991), many sufficient dimension reduction methos have been developed to recover $\mathcal{S}_{Y|\mathbf{x}} = \mathrm{span}(\mathbf{B})$. But all focus on the batch learning setting.
- How to perform dimension reduction on a stochastic data stream has been rarely touched in the literature.

Review of Classic SIR

Denote $\mathbf{\Sigma} = \text{cov}(\mathbf{x}, \mathbf{x}^\top)$. Partition the support of Y into H slices, $I_1, \ldots, I_H$, where $I_h = (q_{h-1}, q_h]$, and $-\infty = q_0 < q_1 < \cdots < q_H = \infty$. Li (1991) suggested to set q_h to be the $(h/H) \times 100\%$-th quantile of Y. Let $p_h = \Pr(Y \in I_h)$ and $\mathbf{m}_h^0 = \mathbf{\Sigma}^{-1}\mathbb{E}\{(\mathbf{x} - \mathbb{E}\mathbf{x}) \mid Y \in I_h\}$. The kernel matrix is defined by

$$\mathbf{M}^0 = \sum_{h=1}^{H} p_h \mathbf{m}_h^0 \mathbf{m}_h^{0\top}.$$

Under mild conditions on $\mathbf{x}$

$$\mathbf{m}_h^0 \in \mathcal{S}_{Y|\mathbf{x}}, \text{ for } h = 1, \ldots, H, \tag{3}$$

This implies that the first K principal eigenvectors of $\mathbf{M}^0$ span $\mathcal{S}_{Y|\mathbf{x}}$ and the $(p - K)$ smallest eigenvalues of $\mathbf{M}^0$ are identically zero.

Online SIR

Here we are concerned with the online learning situation where the observations arrive sequentially as $\{(\mathbf{x}_t, Y_t), t = 1, \ldots\}$ in a data stream. Modifications are needed to suite the online learning.

- Prespecify the slices I_hs . Otherwise sample quantiles woule vary from time to time.
- Define $\mathbf{m}_h = \mathbf{\Sigma}^{-1}\mathbb{E}\{(\mathbf{x} - \mathbb{E}\mathbf{x})\mathbf{1}(Y \in I_h)\}$ and $\mathbf{M} = \sum_{h=1}^{H} \mathbf{m}_h \mathbf{m}_h{}^\top$. The fact that $\mathbf{m}_h^0 = \mathbf{m}_h / p_h$ and p_h is a scalar, together with (3), immediately implies

$$\mathbf{m}_h \in \mathcal{S}_{Y|\mathbf{x}}, \text{ for } h = 1, \ldots, H. \tag{4}$$

Online SIR

The algorithm can be divided into two steps:

- Online update for the kernel matrix
- Online singular value decomposition

Online Update for the Kernel Matrix

We define the sample estimators as follows. Let

$$\widehat{\mathbf{M}}_t = \sum_{h=1}^{H} \widehat{\mathbf{m}}_{t,h}\widehat{\mathbf{m}}_{t,h}^\top = (\widehat{\mathbf{m}}_{t,1}, \ldots, \widehat{\mathbf{m}}_{t,H})(\widehat{\mathbf{m}}_{t,1}, \ldots, \widehat{\mathbf{m}}_{t,H})^\top, \tag{5}$$

$$\widehat{\mathbf{m}}_{t,h} = \widehat{\boldsymbol{\Sigma}}_t^{-1}\left\{\frac{1}{t}\sum_{i=1}^{t}(\mathbf{x}_i - \overline{\mathbf{x}}_t)\mathbf{1}(Y_i \in I_h)\right\}, \widehat{\boldsymbol{\Sigma}}_t = \frac{1}{t}\sum_{i=1}^{t}(\mathbf{x}_i - \overline{\mathbf{x}}_t)(\mathbf{x}_i - \overline{\mathbf{x}}_t)^\top$$

This indicates that updating $\widehat{\mathbf{M}}_{t+1}$ from $\widehat{\mathbf{M}}_t$ amounts to updating $\widehat{\mathbf{m}}_{t+1,h}$ from $\widehat{\mathbf{m}}_{t,h}$.

Online Update for the Kernel Matrix

To update $\widehat{\mathbf{m}}_{t,h}$, we augment $\mathbf{x}$ by adding a row of ones, $\widetilde{\mathbf{x}} = (1, \mathbf{x}^\top)^\top \in \mathbb{R}^{(p+1)\times 1}$. We notice that $\widehat{\mathbf{m}}_{t,h}$ is the least squares estimate of the slope vector by regressing $\{\mathbf{1}(Y_1 \in I_h), \ldots, \mathbf{1}(Y_t \in I_h)\}^\top$ onto $(\widetilde{\mathbf{x}}_1, \ldots, \widetilde{\mathbf{x}}_t)^\top$ linearly. Write $\widetilde{\mathbf{I}} = (\mathbf{0}_{p\times 1}, \mathbf{I}_{p\times p})$ and $\widetilde{\mathbf{y}}_t = \{\mathbf{1}(Y_t \in I_1), \ldots, \mathbf{1}(Y_t \in I_H)\}^\top \in \mathbb{R}^{H\times 1}$. Let

$$\mathbf{A}_t = \sum_{i=1}^{t} \widetilde{\mathbf{x}}_i \widetilde{\mathbf{x}}_i^\top \in \mathbb{R}^{(p+1)\times(p+1)}.$$

It follows immediately that

$$\begin{aligned} (\widehat{\mathbf{m}}_{t,1}, \ldots, \widehat{\mathbf{m}}_{t,H}) &= \widetilde{\mathbf{I}}_{p\times(p+1)} \mathbf{A}_t^{-1} \left(\sum_{i=1}^{t} \widetilde{\mathbf{x}}_i \widetilde{\mathbf{y}}_i^\top \right), \text{ and,} \qquad (6) \\ (\widehat{\mathbf{m}}_{t+1,1}, \ldots, \widehat{\mathbf{m}}_{t+1,H}) &= \widetilde{\mathbf{I}}_{p\times(p+1)} \mathbf{A}_{t+1}^{-1} \left(\sum_{i=1}^{t} \widetilde{\mathbf{x}}_i \widetilde{\mathbf{y}}_i^\top + \widetilde{\mathbf{x}}_{t+1} \widetilde{\mathbf{y}}_{t+1}^\top \right). \end{aligned}$$

Online Singular Value Decomposition via Perturbation

Lemma 1

Let $\mathbf{D} \in \mathbb{R}^{p \times p}$ *be a symmetric matrix and* $(\lambda_j, \mathbf{v}_j)$ *be the eigen-pairs of* $\mathbf{D}$, $j = 1, \ldots, p$. *Assume* $|\lambda_1| > \cdots > |\lambda_d| > \lambda_{d+1} = \cdots = \lambda_p = 0$. *Let* ε *be a very small constant and* $\mathbf{G}$ *be a symmetric matrix. Denote the first order perturbation* $\mathbf{D}(\varepsilon) = \mathbf{D} + \varepsilon \mathbf{G} + O(\varepsilon^2)$ *and the eigen-pairs of* $\mathbf{D}(\varepsilon)$ *by* $\{\lambda_j(\varepsilon), \mathbf{v}_j(\varepsilon)\}$. *Then*

$$\begin{aligned} \lambda_j(\varepsilon) &= \lambda_j + \varepsilon(\mathbf{v}_j^\top \mathbf{G} \mathbf{v}_j) + O(\varepsilon^2), \ \textit{and} \\ \mathbf{v}_j(\varepsilon) &= \mathbf{v}_j + \varepsilon(\lambda_j \mathbf{I}_{p \times p} - \mathbf{D})^+ \mathbf{G} \mathbf{v}_j + O(\varepsilon^2), \ j = 1, \ldots, d, \end{aligned}$$

where $(\lambda_j \mathbf{I}_{p \times p} - \mathbf{D})^+$ *stands for the Moore-Penrose pseudo-inverse of* $(\lambda_j \mathbf{I}_{p \times p} - \mathbf{D})$ *and* $\mathbf{I}_{p \times p}$ *stands for the* $p \times p$ *identity matrix.*

Online Singular Value Decomposition via Perturbation

To put this lemma into online learning, we define

$$\widehat{\mathbf{\Gamma}}_t = \frac{1}{t}\sum_{i=1}^{t}\widehat{\mathbf{M}}_t. \text{ It follows that } \widehat{\mathbf{\Gamma}}_{t+1} = \widehat{\mathbf{\Gamma}}_t - \frac{1}{t+1}(\widehat{\mathbf{\Gamma}}_t - \widehat{\mathbf{M}}_{t+1}). \tag{7}$$

We apply Lemma 1 by defining $\mathbf{D} = \widehat{\mathbf{\Gamma}}_t$, $\mathbf{G} = \widehat{\mathbf{\Gamma}}_t - \widehat{\mathbf{M}}_{t+1}$ and $\varepsilon = -1/(t+1)$.

Algorithm 1: Online SIR via the perturbation

1. Initialize $\widehat{\mathbf{M}}_1$ and $\widehat{\mathbf{\Gamma}}_1$ to obtain $(\widehat{\lambda}_{1,j}, \widehat{\boldsymbol{\beta}}_{1,j})$, $j = 1, \cdots, K$, with a small sample.
2. As the data stream is being collected,
   2.1 online update for the kernel matrix: update $\widehat{\mathbf{M}}_{t+1}$ from $\widehat{\mathbf{M}}_t$ through (6) to obtain

$$\widehat{\mathbf{M}}_{t+1} = \sum_{h=1}^{H} \widehat{\mathbf{m}}_{t+1,h} \widehat{\mathbf{m}}_{t+1,h}^{\top}, \tag{8}$$

   and update $\widehat{\mathbf{\Gamma}}_{t+1}$ from $\widehat{\mathbf{\Gamma}}_t$ through (7);

   2.2 online singular value decomposition: update the eigen-pairs of $\widehat{\mathbf{\Gamma}}_{t+1}$ through

$$\widehat{\lambda}_{t+1,j} = \widehat{\lambda}_{t,j} - (t+1)^{-1} \widehat{\boldsymbol{\beta}}_{t,j}^{\top} (\widehat{\mathbf{\Gamma}}_t - \widehat{\mathbf{M}}_{t+1}) \widehat{\boldsymbol{\beta}}_{t,j}, \text{ and} \tag{9}$$

$$\widehat{\boldsymbol{\beta}}_{t+1,j} = \widehat{\boldsymbol{\beta}}_{t,j} - (t+1)^{-1} (\widehat{\lambda}_{t,j} \mathbf{I}_{p \times p} - \widehat{\mathbf{\Gamma}}_t)^{+} (\widehat{\mathbf{\Gamma}}_t - \mathbf{M}_{t+1}) \widehat{\boldsymbol{\beta}}_{t,j}. \tag{10}$$

Theoretical Property of Algorithm 1

(C1) The smallest eigenvalue of $\mathbf{\Sigma}$ is bounded away from zero.

(C2) The observations $\{(\mathbf{x}_t, Y_t), t = 1, \ldots\}$ are independent and identically distributed.

(C3) The nonzero eigenvalues of $\mathbf{M}$ are all distinct.

Theorem 2

Under conditions (C1)-(C3), the eigen-pair $(\widehat{\lambda}_{t,j}, \widehat{\boldsymbol{\beta}}_{t,j})$ *of* $\widehat{\mathbf{\Gamma}}_t$ *generated by Algorithm 1 converges almost surely as* $t \to \infty$*, for* $j = 1, \ldots, K$*.*

Online Singular Value Decomposition via SGD

The stochastic gradient descent optimization was also widely used for online analysis of independent data streams. We notice that, seeking the K principal eigenvectors of $\widehat{\mathbf{M}}_{t+1}$ amounts to

$$\begin{aligned} \underset{\mathbf{B}}{\text{maxmize}} \quad & \text{tr}(\widehat{\mathbf{M}}_{t+1}\mathbf{B}\mathbf{B}^\top) \\ \text{subject to} \quad & \mathbf{B}^\top\mathbf{B} = \mathbf{I}_{K\times K} \end{aligned} \tag{11}$$

The first order derivative of the objective function is $2\widehat{\mathbf{M}}_{t+1}\mathbf{B}$, which inspires the proposal of the following online stochastic gradient descent optimization:

$$\widehat{\mathbf{B}}_{t+1} = \mathcal{P}_{\text{orth}}(\widehat{\mathbf{B}}_t + \gamma_t\widehat{\mathbf{M}}_{t+1}\widehat{\mathbf{B}}_t), \tag{12}$$

where γ_t is a small step-size.

Algorithm 2: Online SIR via the SGD

1. Initialize $\widehat{\mathbf{M}}_1$ to obtain $\widehat{\mathbf{B}}_1$ with a small sample.
2. As the data stream is being collected,
   2.1 online update for the kernel matrix: update $\widehat{\mathbf{M}}_{t+1}$ from $\widehat{\mathbf{M}}_t$ through (8),
   2.2 online singular value decomposition: update the eigenspace of $\widehat{\mathbf{M}}_{t+1}$ through

$$\widehat{\mathbf{B}}_{t+1} = \mathcal{P}_{\mathrm{orth}}(\widehat{\mathbf{B}}_t + \gamma_t \widehat{\mathbf{M}}_{t+1} \widehat{\mathbf{B}}_t), \tag{13}$$

Theoretical Property of Algorithm 2

(C4) Let $\gamma_t = t^{-1}$, $t = 1, 2, \ldots$.

Theorem 3

Under conditions (C2)-(C4), the column space of $\widehat{\mathbf{B}}_t = (\widehat{\boldsymbol{\beta}}_{t,1}, \ldots, \widehat{\boldsymbol{\beta}}_{t,K})$ *converges almost surely to the column space of* $\mathbf{M}$*, as* $t \to \infty$*.*

The column space of $\mathbf{M}$, defined as the space spanned by the eigenvectors associated with nonzero eigenvalues, coincides the central subspace $\mathcal{S}_{Y|\mathbf{X}}$, as demonstrated in (4). Therefore, $\widehat{\mathbf{B}}_t$ is an asymptotically consistent basis matrix of $\mathcal{S}_{Y|\mathbf{X}}$.

Numerical Validation

Table 1: Brief description of several UCI benchmark datasets

dataset	task	training	test	p	K
wbc	classification	524	175	9	1
wdbc	classification	426	143	30	1
wpbc	classification	148	50	33	1
spambase	classification	3450	1151	57	1
magic04	classification	14265	4755	10	1
krvskp	classification	2397	799	36	1
digits069	classification	2219	1035	16	2
digits	classification	7494	3498	16	8
housing	Regression	379	127	14	2
airfoil	Regression	1127	376	6	1

Table 2: The prediction errors of three online learners and batch learning methods ("full batch", "batch SIR", "perturbed SIR" and "SIR with SGD"). We use classification tree and report the mis-classification error rates in classification problems, and fit regression trees and report the R square in regression problems.

dataset	full batch	batch SIR	perturbed SIR	SIR with SGD
wbc	0.0591	0.0302	0.0391	0.0343
wdbc	0.0749	0.0524	0.0438	0.0392
wpbc	0.2772	0.2384	0.2536	0.2316
spambase	0.1059	0.1003	0.112	0.093
magic04	0.1784	0.2072	0.2296	0.2064
krvskp	0.0354	0.0765	0.0936	0.0568
digits069	0.0377	0.0454	0.0203	0.0251
digits	0.2613	0.2347	0.2433	0.2541
housing	0.4249	0.6971	0.4953	0.4652
airfoil	0.2839	0.3506	0.4108	0.3678

Table 3: The mis-classification error rates for the "digits" dataset.

K	batch SIR	perturbed SIR	SIR with SGD
1	0.7344	0.7684	0.7413
2	0.5986	0.6512	0.5638
3	0.4786	0.5266	0.5266
4	0.3808	0.4677	0.3756
5	0.3908	0.3445	0.3917
6	0.2799	0.2976	0.3625
7	0.2762	0.2882	0.3139
8	0.2348	0.2433	0.2839
9	0.2270	0.2333	0.2627

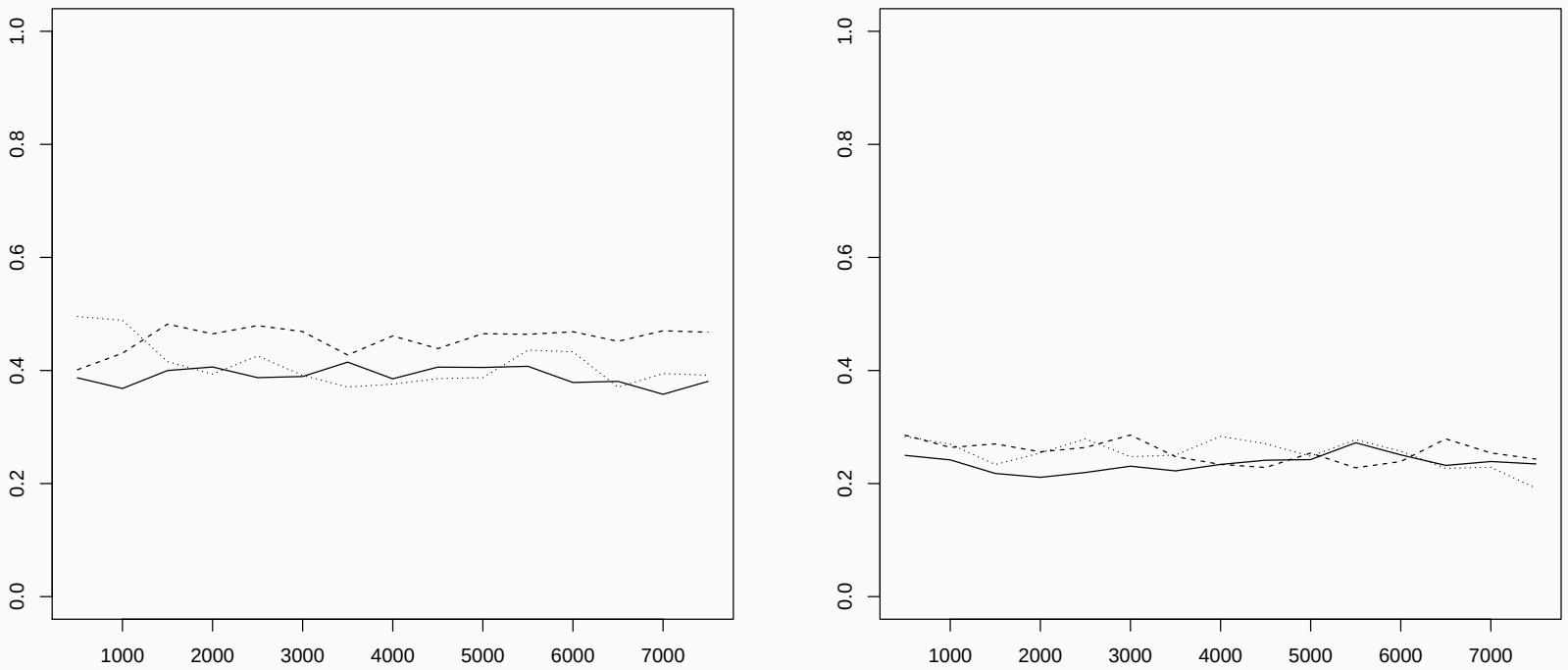


Figure 1: The horizontal axis stands for the sample size used by three dimension reductions approaches: "batch SIR", "perturbed SIR" and "SIR with SGD"; and the vertical axis stands for the mis-classification error rates. The reduced dimension $K = 4$ in the left and $K = 8$ in the right plot. The solid line, dashed line, and doted line represent batch SIR, perturbed SIR and SIR with SGD, respectively.

Concluding Remarks

In this paper we introduce a modified two-step online sliced inverse regression procedure.

- Online update for the kernel matrix
- Online update for the singular value decomposition
  - Perturbation method
  - Stochastic gradient descent method

Our proposed approach can be readily generalized to other sufficient dimension reduction methods such as sliced average variance estimation and directional regression. Investigations along this way are warranted.

Thanks!

References

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